



AMTEC

AGRICULTURAL MACHINERY TESTING AND EVALUATION CENTER

TEST REPORT NO. 2013 – 74



**SCANCON Rice Hull-Fed Furnace
Model: SA 32**

SCANCON SA 32 RICE HULL-FED FURNACE

Test Report No.	:	2013 - 74
Date of Test	:	May 21, 2013
Test Valid Until	:	May 21, 2018
Brand	:	SCANCON
Model	:	SA 32
Type	:	Indirect-fired, boiler-type, rice hull-fed
Manufacturer	:	SCANCON, INC. 35 Sct. Torillo Quezon City
Test Requested by:	:	Manufacturer
Machine Classification	:	Commercial

SUMMARY

The SCANCON SA 32 Rice Hull-Fed Furnace was tested for performance using rice husk with 2.48%_{w.b.} Moisture Content as fuel. It had a Burning Efficiency of 82.8% and a Furnace Efficiency of 11.9%. The Fuel Consumption of this furnace using rice husk was 232.0 kg/h while the Total Power Consumption of the electric motors which drove the components of the furnace was 2.99 kW with Total Load Current of 14.7 amperes and Line Voltage of 215 volts.

One person was needed to operate the furnace.

PURPOSE AND SCOPE OF TEST

The SCANCON SA 32 Rice Hull-Fed Furnace was tested to determine its operating performance. The test was conducted to verify and assess the following:

- 1.0 Burning Efficiency
- 2.0 Furnace Efficiency
- 3.0 Furnace Fuel Consumption Rate
- 4.0 Power Consumption

PLACE AND DATE OF TEST

The SCANCON SA 32 Rice Hull-Fed Furnace was tested in Brgy. Calaocan, San Jose City on May 21, 2013.

METHODS OF TEST

The SCANCON SA 32 Rice Hull Furnace was tested based on the “PAES 243:2010 Agricultural Machinery – Biomass Furnace - Methods of Test”.

OPERATOR’S MANUAL

No operator’s manual was submitted.

DESCRIPTION OF THE FURNACE

The SCANCON SA 32 Rice Hull-Fed Furnace (**Figure 1**) was basically an indirect-fired, boiler-type rice hull-fed furnace. It consisted of the following components: rice hull tank, rice hull feeder, blower which served as fuel injector, combustion chamber, scraper system, boiler pipes, circulation pump, flue gas exhauster, heat exchangers, expansion tank, ash collector and control panel.

Rice hulls from the fuel tank were fed to the furnace through a vibrator (**Figure 2a**) which was installed beneath the fuel tank and a fan (**Figure 2b**) which injects the rice hull into the combustion chamber. An oscillating scraper system (**Figure 3**) which agitates the fire bed and aid in the discharge of ash was installed inside the combustion chamber. There were 83 pieces of Schedule 80 boiler tubes (**Figure 4**) inside the furnace where flue gas entered and was sucked by a fan (**Figure 5**) which served as flue gas exhauster.

This biomass boiler was designed to use rice husk as fuel. Hot water produced by this boiler was pumped through a circulation pump (**Figure 6**) to two heat exchangers (**Figure 7**) mounted at the suction side of two 10-ton recirculating dryers.

A control panel (**Figure 8**) which housed the volt meter and switches for the thermostat and motors for the rotating components of the furnace was installed.



Figure 1. SCANCON Rice Hull-Fed Furnace
Model: SA 32

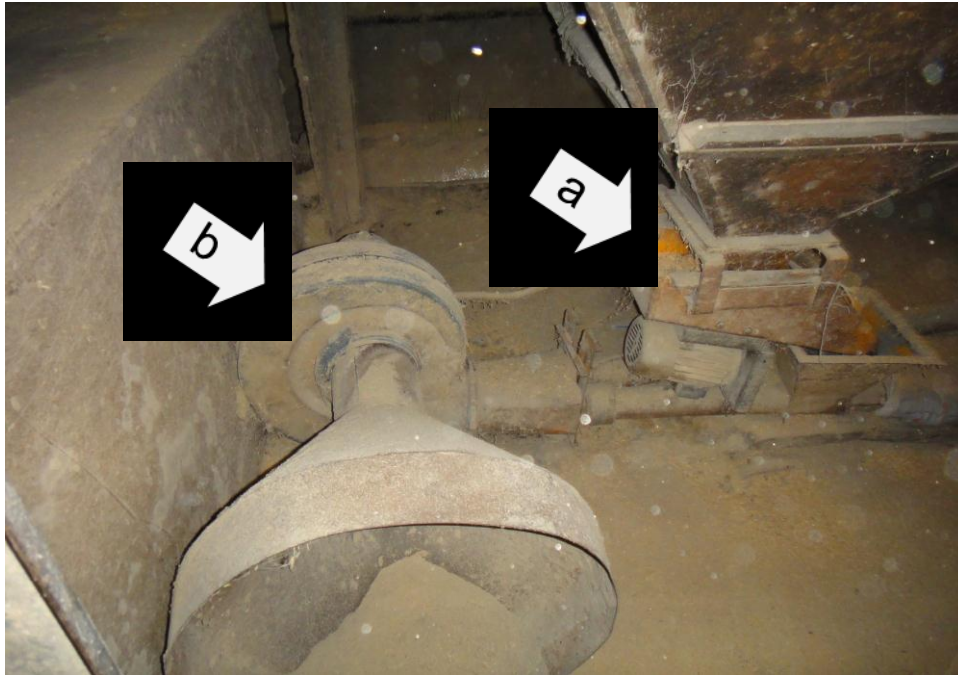


Figure 2. a) Vibrator
b) Fan as fuel injector



Figure 3. Scraper system

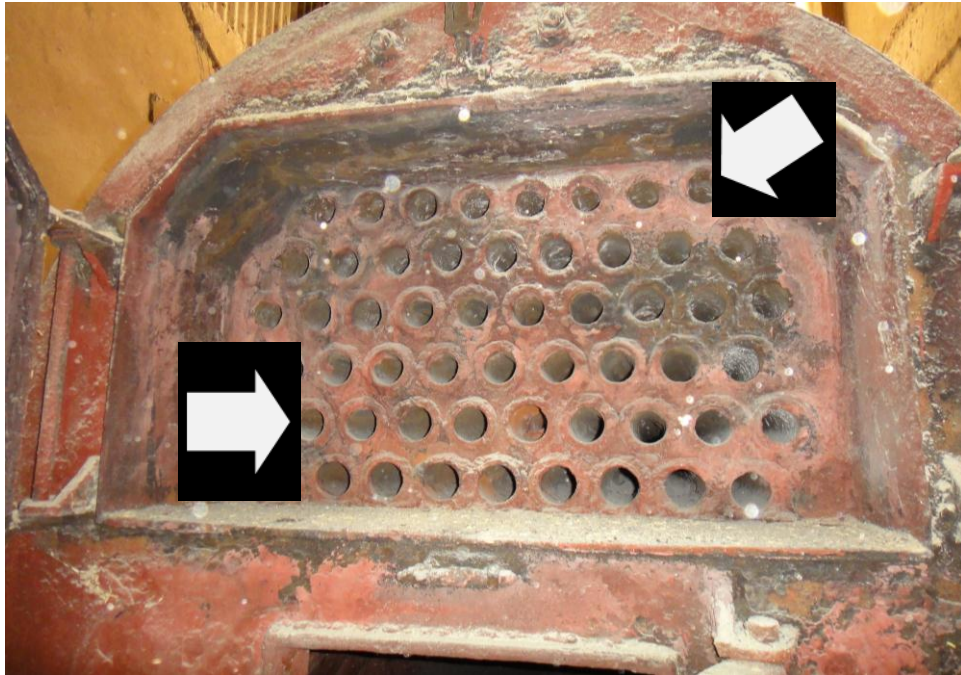


Figure 4. Boiler tubes

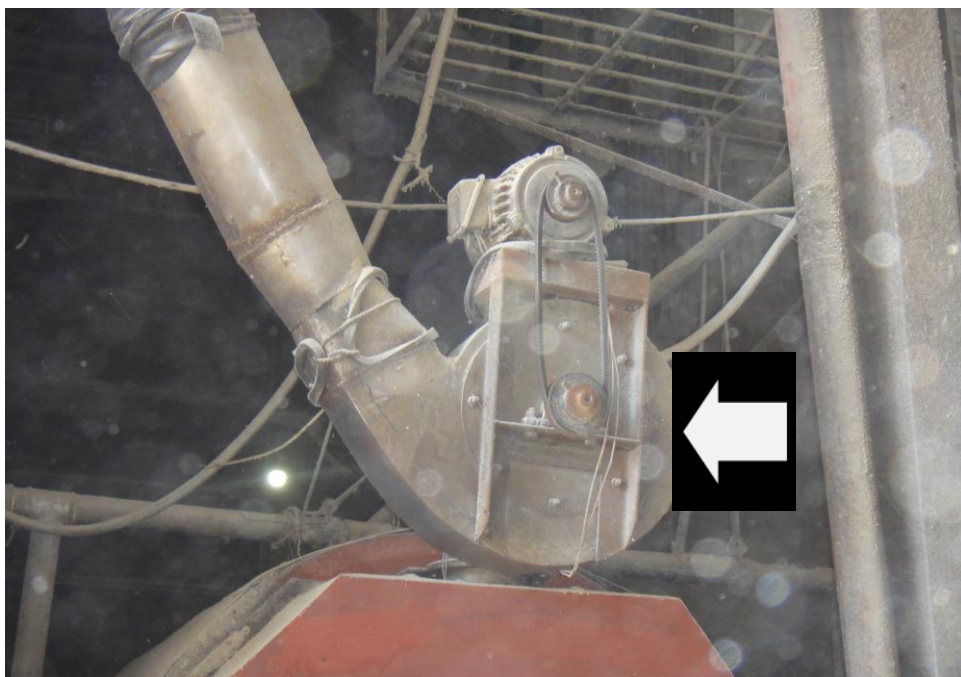


Figure 5. Aspirating fan for flue gas

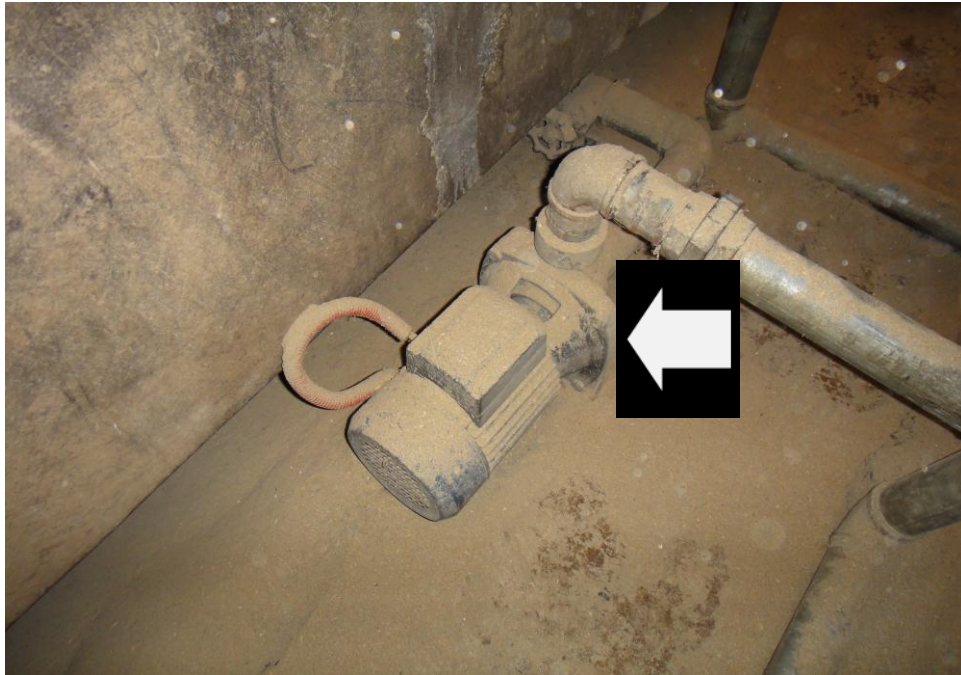


Figure 6. Circulation pump

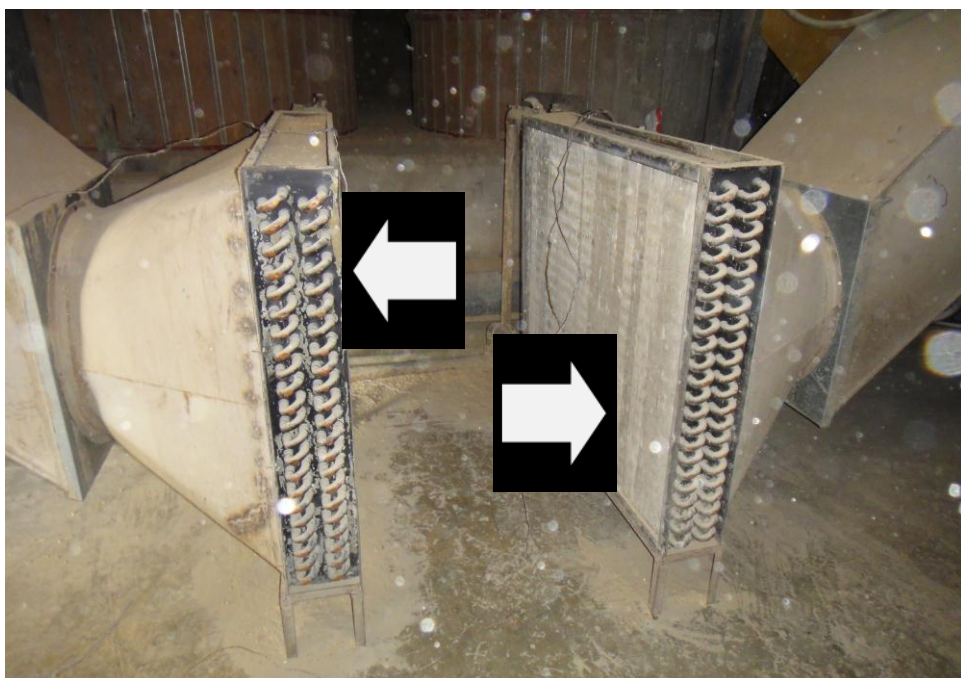


Figure 7. Heat Exchangers



Figure 9. Control panel

RESULTS AND DISCUSSION

The SCANCON SA 32 Rice Hull-Fed Furnace was tested for performance.

Firing of Furnace

Piece of paper ignited with a lighter was used to fire the furnace (**Figure 10**). It took 0.92 hour and 200 kilograms of rice husk before the heat produced by the furnace reached stable temperature.

Burning Efficiency

The burning efficiency of the furnace was the percentage of heat released by the fuel inside the furnace based on the gross heating value of the fuel. Using rice husk with an average moisture content of 2.48%_{W.B.}, the furnace had a Burning Efficiency of 82.8%.

Furnace Efficiency

The furnace efficiency was the ratio of the heat supplied by the furnace to the heat released by the fuel, expressed in percent. The Furnace Efficiency was 11.9 %.

Furnace's Fuel Consumption

The fuel consumption of the rice hull-fed furnace was determined by getting the total amount of rice hull consumed during the test divided by the total time of operation. The Fuel Consumption of the furnace was 232.0 kg/h of rice husk.

Power Consumption

The Total Power Consumption of the electric motors which drove the components of the furnace was 2.99 kW with Total Load Current of 14.7 amperes and Line Voltage of 215 volts.

Labor Requirement

One person (**Figure 11**) was needed to fire and operate the furnace during the actual drying operation.

Observation

The furnace operated smoothly and no breakdown or failure of components occurred during the test.

The following was also observed:

1. The furnace was provided with thermostat control for uniform temperature output.



Figure 10. Firing of furnace



Figure 11. Operator of the furnace

PERFORMANCE TEST

Furnace Tested : SCANCON Rice Hull-Fed Furnace
Model: SA 32

Location of Test: Brgy. Calaocan
San Jose City

Date of Test : May 21, 2013

Table 1. Condition of rice husk (fuel) used during the test

ITEMS	Data
1. Moisture Content (%)	2.48
2. Bulk Density (kg/m ³)	157.4

Table 2. Speed of rotating components (with load) during the test

ITEMS	Data
1. Gear Reducer Shaft (for scraper system), rpm	1223
2. Scraper System Eccentric Shaft (rpm)	20
3. Discharge Ash Crumbler Shaft (rpm)	294
4. Aspiration Fan for Flue Gas (rpm)	1407

Table 3. Results of Performance Test of SCANCON SA 32 Rice Hull-Fed Furnace

ITEMS	DATA
1. Furnace Firing Time (h)	0.92
2. Fuel Consumed During Firing (kg)	200.0
3. Ambient Air Temperature (°C)	
a) Dry Bulb	33.3
b) Wet Bulb	25.6
4. Ambient Air Relative Humidity (%)	54.1
5. Average Furnace Output Heated Air Temperature (°C)	58.9
6. Fan Air Velocity (m/s)	4.40
7. Air Flow Rate (m ³ /min)	216.5
8. Rice Husk Consumed (kg)	406.0
9. Total Operating Time (h)	1.75
10. Rice Husk Consumption Rate (kg/h)	232.0
11. Power Consumption	
a) Power, kW	2.99
b) Voltage, V	215
c) Current, A	14.7
12. Burning Efficiency (%)	82.8
13. Furnace Efficiency (%)	11.9

Table 4. General information of fuel samples and ash samples brought to AMTEC laboratory for analysis

ITEMS	RICE HUSK	DISCHARGE ASH
1. Initial dry weight (g)	4.06	2.40
2. Residue weight (g)	0.83	1.44
3. Ash (%)	20.44	-
4. Ash Residue (%)	-	34.13

ANNEX 1. FORMULA USED IN THE COMPUTATION OF HEATING PARAMETERS

$$1. \text{ Burning Efficiency (\%)} = \frac{\text{Heat released by the fuel (KJ/kg)}}{\text{Gross heating value of fuel (KJ/kg)}} \times 100$$

Volatilized Method:

$$\text{Burning Efficiency (\%)} = \frac{100 - A_r}{100 - A} \times 100$$

Where:

A = Percentage of ash (dry basis) in fuel sample after complete combustion in an electric furnace

$$= \frac{\text{Residue Weight (rice hull)}}{\text{Initial Dry Weight (rice hull)}} \times 100$$

A_r = Percentage of ash residue (dry basis) in the sample collected from the furnace

$$= \left[\frac{\text{Initial Dry Wt. (Discharge Ash)} - \text{Residue Wt. (Discharge Ash)}}{\text{Residue Wt. (Discharge Ash)}} + 1 \right] \times \% \text{Ash}$$

$$2. \text{ Furnace Efficiency (\%)} = \frac{\text{Heat supplied by the furnace}}{\text{Heat released by the fuel}} \times 100$$

Where,

Heat supplied by the furnace, Q_s (kJ/h)

$$Q_s = (\text{mass flow rate of heated air, kg/h}) \times (\text{specific heat of heated air, kJ/kg}) \times (\text{Drying air temperature, } ^\circ\text{C} - \text{Ambient Temperature, } ^\circ\text{C})$$

Where:

$$\text{Mass flow rate of heated air} = (\text{volume flow rate of heated air}) \times (\text{density of heated air})$$

Heat released by the fuel, Q_r (kJ/h)

$$Q_r = (\text{rice hull consumed, kg/h}) \times (\text{gross heating value of fuel, kJ/kg})$$

3. Heating System Efficiency, E_{HS} (%)

$$E_{HS} = \frac{Q_s}{Q_f} \times 100$$

Where,

E_{HS} = Heating system efficiency (%)

Q_s = Heat supplied to the dryer (kJ/h)

Q_f = Heat available in the fuel (kJ/h)

To solve for Q_s :

$$Q_s = \frac{\Delta h \times Q}{v} \times 60$$

Where,

Δh = Change in enthalpy (kJ/kg d.a.)

Q = Air flow rate (m³/min)

v = Specific volume of fresh air (m³/kg d.a.)

To solve for Q_f :

$$Q_f = F_c \times H_{vf}$$

Where,

F_c = Fuel consumption (kg/h)

H_{vf} = Heating value of the fuel used (kJ/kg)

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Date

N.B.

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